A comparative analysis of Von-Heijne, SVM and Perceptron Models for the detecton of Signal Peptides.

Abstract

Motivation: This is the motivation

Results: This are the results.

Introduction

For a protein to enter the secretory pathway, in both eukaryotic and prokaryotic cells, it must be endowed with a specific target signal. Often, this signal takes the shape of a short sequences located at the N-terminus of proteins [http://www.ncbi.nlm.nih.gov/pubmed/2197415]. The signal peptide has a distinct three-domain structure, depicted in Figure 1, with a positively charged N-terminal region (n-region), a central hydrophobic region (h-region) and a more polar C-terminal region (c-region) containing the cleavage site [http://www.ncbi.nlm.nih.gov/pubmed/2197415]. Given their importance in many aspects of cell biology, the accurate detection of SPs is a crucial task in bioinformatics, which has been tackled before with a variety of approaches, including machine learning (SVMs [https://pubmed.ncbi.nlm.nih.gov/19470175], [https://pubmed.ncbi.nlm.nih.gov/21959131/] and Bayesian networks [http://www.ncbi.nlm.nih.gov/pubmed/18989393]) and deep learning [https://academic.oup.com/bioinformatics/article/34/10/1690/4769493?login=false].

Figure 1: Signal peptide structure

With this work, we propose a comparative analysis of three different approaches for the detection of SPs, namely the Von-Heijne model, a Support Vector Machine (SVM) and a Multi-layer Perceptron (MLP).

Materials and Methods

Data Collection

Protein sequences were queried from UniprotKB, and collected into 2 distinct sets, positive and negative. For Positive sequences, we selected sequences having experimental evidence for the presence of a signal peptide (`ft_signal_exp:*`) For the negative set, proteins not annotated with a signal peptide and endowed with experimental evidence for subcellular localization in non-secretory comparments, namely nucleus (`cc_scl_term_exp:SL-0191`), peroxisome (`cc_scl_term_exp:SL-0204`), cell membrane / plasma membrane (`cc_scl_term_exp:SL-0039`), cytosol (`cc_scl_term_exp:SL-0091`), plastid (`cc_scl_term_exp:SL-0209`) or mitochondrion (`cc_scl_term_exp:SL-0173`).

Both sets were also limited to the taxa of eukaroyot and areviewed entries, and non-fragment nature.

Further selection on the positive entries was made to select proteins which had informaton about the cleavage site and a Signal Peptide with a length of at least 14bp.

Dataset	**Before Clustering**	**After Clustering**
Positive	2942	1093
Negative (total)	20806	9028
Negative (w/o TM)	18301	-
Negative (with TM)	2505	-

Table 1: Dataset sizes before and after clustering

For the negative set, additional information was added to the data representing the presence of a transmembrane helix, whose hydrophobicity profile mimicks the one of a signal peptide.

Von-Heijne Model

The full dataset was split into a training and a test set with a ratio of 0.8. A Position-Specific Weight Matrix (PSWM) was then generated using 16bp windows (13bp upstream to 2bp downstream) surrounding the cleavage site. Amino acid frequencies were normalized using background amino acid frequencies from SwissProt. After initializing the matrix with pseudocounts of 1, the weights are computed as follows:

$$w(a, i) = \log_2\left(\frac{f(a, i)}{f(a)}\right)$$

Where $f(a, i)$ is the frequency of amino acid a at position i in the training set, and $f(a)$ is the background frequency of amino acid a in SwissProt. The score for a given sequence is then computed as:

$$S = \max_{j=1}^{90-16+1} \sum_{i=1}^{16} w(a_{j+i-1}, i)$$

scores are computed over the first 90bp of the given sequence, using a sliding window 16bp long. Then, a sequence was labeled as positively endowed with a SP if the Score S is greater than a threshold t.

$$\text{label} = \begin{cases} \text{positive} & \text{if } S > t \\ \text{negative} & \text{if } S \leq t \end{cases}$$

The threshold was computed using 5-fold cross validation.

Clustering

Clustering was performed using mmseqs2 [cit], using parameters =`c 0.4 --min-seq-id 0.3 --cov-mode 0 --cluster-mode 1`, giving clusters with less than 30% pairwise identity.

Results

For the Von-Hejne model the computed optimal threshold was 9.25.